

Ex. No. 3	ADD INTERACTIVITY TO APP – INPUT AND FORMS
Date of Exercise	
GitHub Host Link	https://github.com/JenithB/flutter.git

Aim

To create an interactive Flutter student registration form with input fields, form validation, and a submit button.

Description

This experiment demonstrates how to create an interactive form in Flutter using `Form`, `TextFormField`, `GlobalKey<FormState>`, and `ElevatedButton`. The form accepts student name, email, and password. Validation messages are displayed when required fields are left empty.

Procedure:

1. Create a new Flutter project using the command:

```
flutter create student_form
```

2. Open the project folder:

```
cd student_form  
code.
```

3. Open the lib/main.dart file.
4. Create a `GlobalKey<FormState>` to manage form validation.
5. Create a `Form` widget and assign the form key.
6. Add a `TextFormField` for entering the student name.
7. Add an email input field using `TextInputType.emailAddress`.
8. Add a password field using `obscureText: true`.
9. Add an `ElevatedButton` for submitting the form.
10. Use the `validate()` method to check the entered information.
11. Display a success message using `SnackBar` when the form is valid.
12. Run the application and test the validation by clicking Submit without entering any information.

Program

```
import 'package:flutter/material.dart';

void main() {
  runApp(const StudentFormApp());
}

class StudentFormApp extends StatelessWidget {
  const StudentFormApp({super.key});

  @override
  Widget build(BuildContext context) {
    return MaterialApp(
      debugShowCheckedModeBanner: false,
      title: 'Student Registration',
      theme: ThemeData(
        primarySwatch: Colors.blue,
      ),
      home: const StudentForm(),
    );
  }
}

class StudentForm extends StatefulWidget {
  const StudentForm({super.key});

  @override
  State<StudentForm> createState() => _StudentFormState();
}

class _StudentFormState extends State<StudentForm> {
  final _formKey = GlobalKey<FormState>();

  @override
  Widget build(BuildContext context) {
    return Scaffold(
      appBar: AppBar(
        title: const Text('Student Registration'),
        backgroundColor: Colors.blue,
        foregroundColor: Colors.white,
      ),

      body: Padding(
        padding: const EdgeInsets.all(20.0),

        child: Form(
          key: _formKey,

          child: Column(
            crossAxisAlignment: CrossAxisAlignment.stretch,
```

children: [

// Student Name

```
TextFormField(  
  decoration: const InputDecoration(  
    labelText: 'Student Name',  
    border: OutlineInputBorder(),  
  ),
```

```
  validator: (value) {  
    if (value == null || value.isEmpty) {  
      return 'Enter student name';  
    }  
    return null;  
  },  
)
```

const SizedBox(height: 15),

// Email

```
TextFormField(  
  keyboardType: TextInputType.emailAddress,
```

```
  decoration: const InputDecoration(  
    labelText: 'Email',  
    border: OutlineInputBorder(),  
  ),
```

```
  validator: (value) {  
    if (value == null || value.isEmpty) {  
      return 'Enter email';  
    }  
  }
```

```
  if (!value.contains('@')) {  
    return 'Enter a valid email';  
  }
```

```
  return null;  
},  
)
```

const SizedBox(height: 15),

// Password

```
TextFormField(  
  obscureText: true,
```

```
  decoration: const InputDecoration(  
    labelText: 'Password',  
    border: OutlineInputBorder(),  
  ),
```

```
    validator: (value) {
      if (value == null || value.isEmpty) {
        return 'Enter password';
      }

      if (value.length < 6) {
        return 'Password must be at least 6 characters';
      }

      return null;
    },
  ),

  const SizedBox(height: 20),

  // Submit Button
  ElevatedButton(
    onPressed: () {
      if (_formKey.currentState!.validate()) {
        ScaffoldMessenger.of(context).showSnackBar(
          const SnackBar(
            content: Text(
              'Form submitted successfully',
            ),
          ),
        );
      }
    },

    child: const Text('Submit'),
  ),
],
),
),
),
);
}
}
```

Output

Student Registration

Register Number

URK23CS1268

Student Name

Jenith B

Email

Password

Submit

The screenshot shows an IDE with the following components:

- EXPLORER:** A file tree on the left showing a project structure with folders like 'student_form_app' and 'lib', and files like 'main.dart'.
- main.dart:** The main code file is open, showing Dart code for a Flutter app. It includes imports, a main function, and two widget classes: `StudentFormApp` (a `StatelessWidget`) and `StudentForm` (a `StatefulWidget`). The `StudentForm` class has a `createState` method that returns a `StudentFormState` object.
- TERMINAL:** The bottom panel shows the output of a `flutter run` command. The output indicates that the command should be run from the root of the Flutter project.

Result

Thus, an interactive Student Registration Form was successfully created using Flutter. The application contains Name, Email, Password, Validation, and Submit Button, and displays appropriate validation messages for empty fields.

